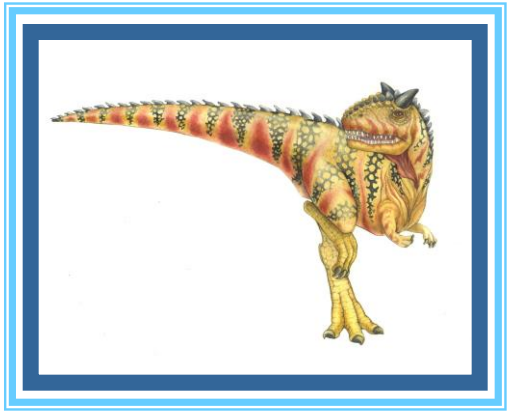
Chapter 8: Main Memory



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Chapter 8: Memory Management

- Contiguous Memory Allocation
- Multiple-Partition Allocation
- Fragmentation
- Segmentation
- Paging



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Contiguous Allocation

- Main memory must support both OS and user processes.
- Limited resource, must allocate efficiently.
- Contiguous allocation is one early method.

- Main memory usually has two **partitions**:
  - Resident operating system, usually held in low memory with interrupt vector.
  - User processes then held in high memory.
  - Each process contained in single contiguous section of memory.



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Contiguous Allocation (Cont.)

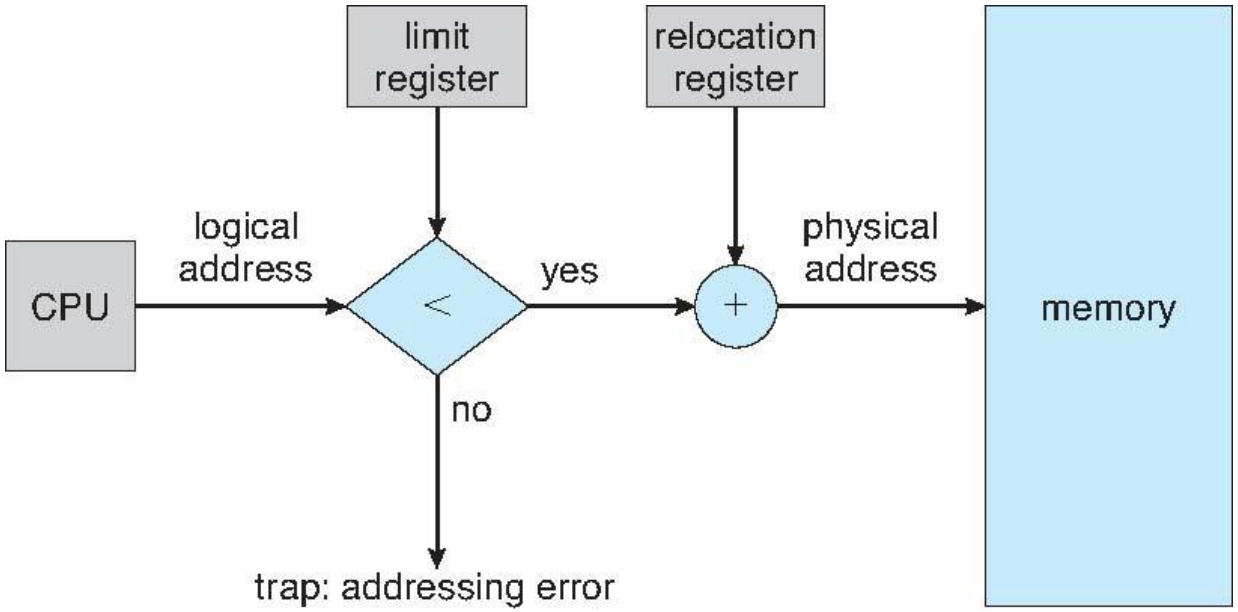
- Relocation registers used to protect user processes from each other, and from changing operating-system code and data
  - Base register contains value of smallest physical address.
  - Limit register contains range of logical addresses – each logical address must be less than the limit register.

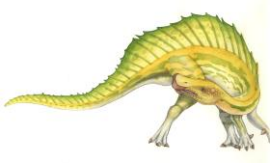


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Hardware Support for Relocation and Limit Registers





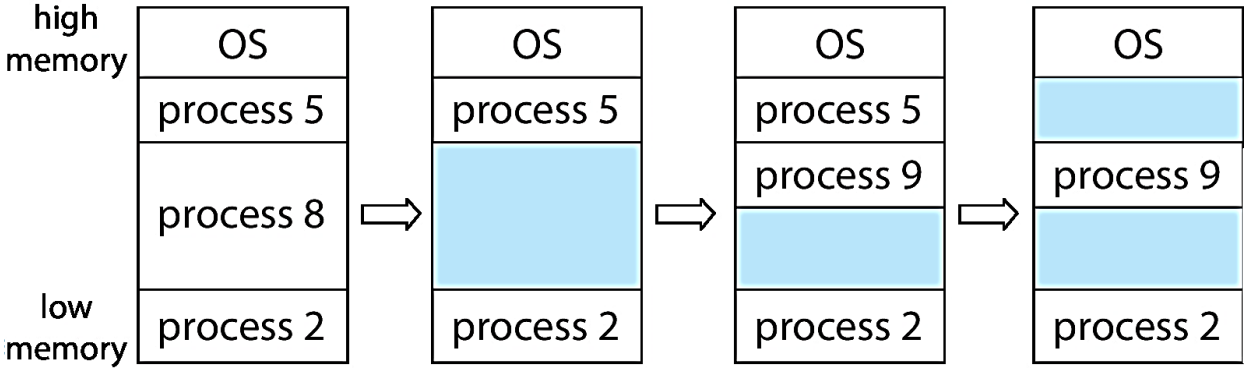
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Multiple-Partition Allocation

- Degree of multiprogramming limited by number of partitions.
- **Variable-partition** sizes for efficiency (sized to a given process' needs).
- **Hole** – block of available memory; holes of various size are scattered throughout memory.
- When a process arrives, it is allocated memory from a hole large enough to accommodate it.
- Process exiting frees its partition, adjacent free partitions combined.
- Operating system maintains information about:

  a) allocated partitions b) free partitions (hole)





Dynamic Storage-Allocation Problem

How to satisfy a request of size ***n*** from a list of free holes?

- **First-fit**: Allocate the ***first*** hole that is big enough
- **Best-fit**: Allocate the ***smallest*** hole that is big enough; must search entire list, unless ordered by size
  - Produces the smallest leftover hole
- **Worst-fit**: Allocate the ***largest*** hole; must also search entire list
  - Produces the largest leftover hole

First-fit and best-fit better than worst-fit in terms of speed and storage utilization



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Fragmentation

- **External Fragmentation** – total memory space exists to satisfy a request, but it is not contiguous.
- **Internal Fragmentation** – allocated memory may be slightly larger than requested memory; this size difference is memory internal to a partition, but not being used.
- Reduce external fragmentation by **compaction**
  - Shuffle memory contents to place all free memory together in one large block.



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Segmentation

- Segmentation is a memory-management scheme that supports this programmer view of memory.
- Programmers prefer to view memory as a collection of variable-sized segments, with no necessary ordering among the segments
- A logical address space is a collection of segments.
- Each segment has a name and a length.
- The addresses specify both the segment name and the offset within the segment.



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Segmentation

- The programmer therefore specifies each address by two quantities: a segment name and an offset.
- Thus, a logical address consists of a two tuple:

  < segment number, offset>

- Normally, when a program is compiled, the compiler automatically constructs segments reflecting the input program.

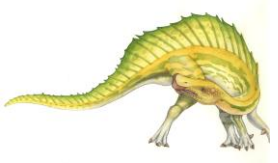


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Segmentation

- A C compiler might create separate segments for the following:
  1. The code
  2. Global variables
  3. The heap, from which memory is allocated
  4. The stacks used by each thread
  5. The standard C library



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Segmentation Architecture

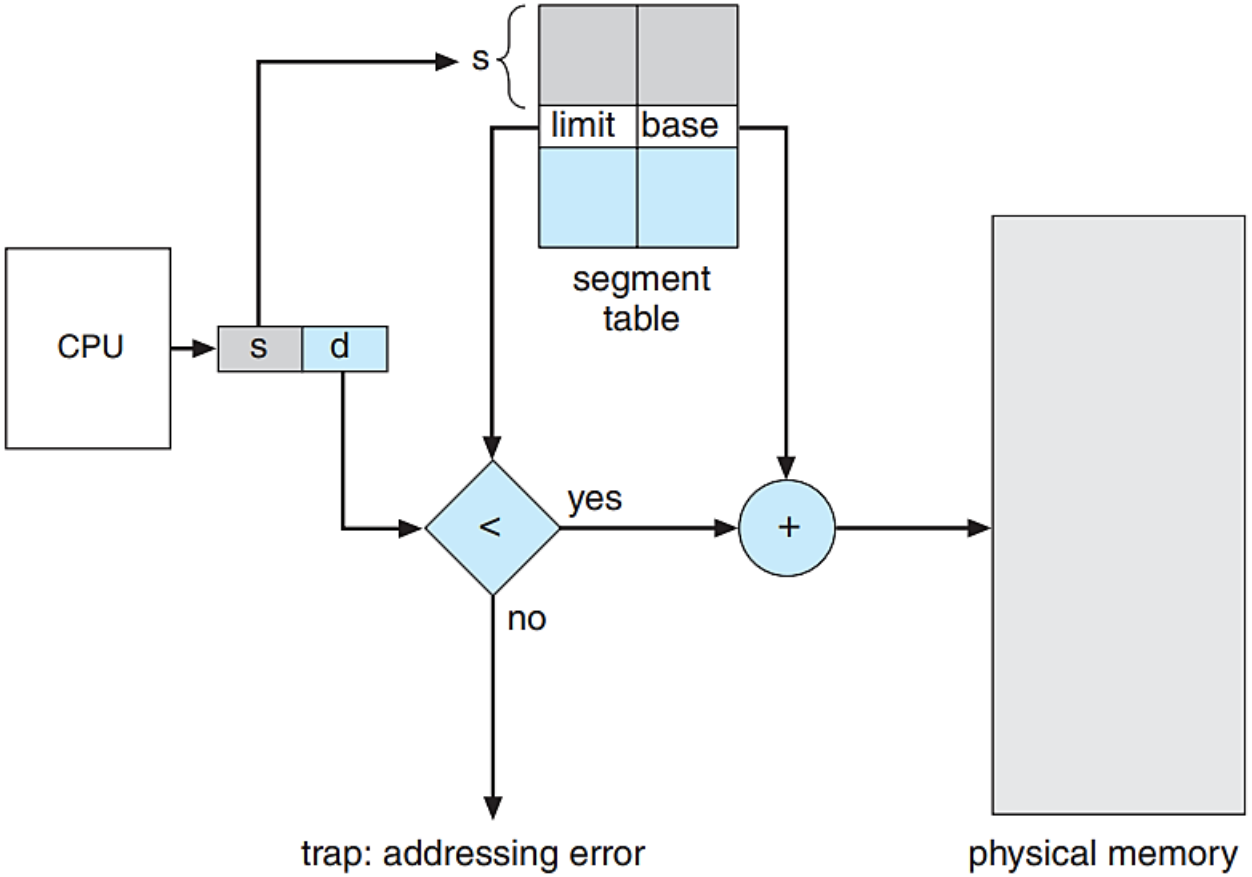
- Segment table maps two dimensional physical addresses, each table entry has:
  - **Segment base:** contains the starting physical address where the segment resides in memory.
  - **Segment limit:** specifies the length of the segment.



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Segmentation Architecture

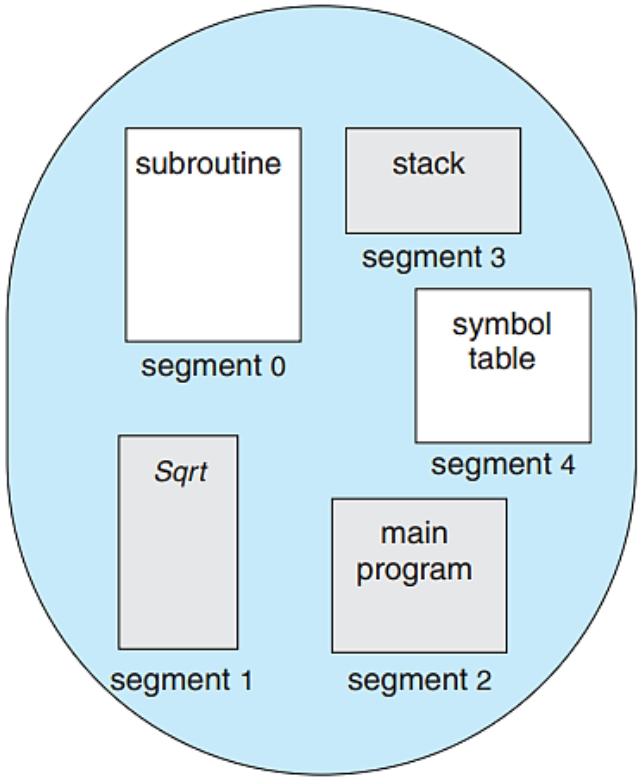




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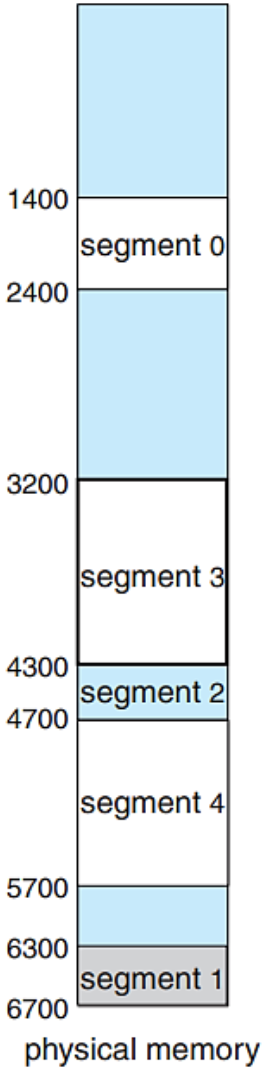


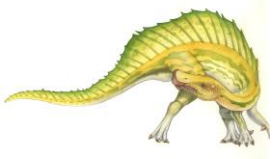
Example of Segmentation



	limit	base
0	1000	1400
1	400	6300
2	400	4300
3	1100	3200
4	1000	4700

segment table





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Paging

- Segmentation permits the physical address space of a process can be noncontiguous.
- Paging is another memory-management scheme that offers this advantage.
- However, paging avoids external fragmentation and the need for compaction, whereas segmentation does not.
- Most memory-management schemes used before the introduction of paging suffered from this problem.



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Paging

- The basic method for implementing paging involves:
  - Breaking physical memory into fixed-sized blocks called **frames.**
  - Breaking logical memory into blocks of the same size called **pages**.
- To run a program of size ***N*** pages, need to find ***N*** free frames and load program.
- Set up a **page table** to translate logical to physical addresses.
- Still have Internal fragmentation.



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Address Translation Scheme

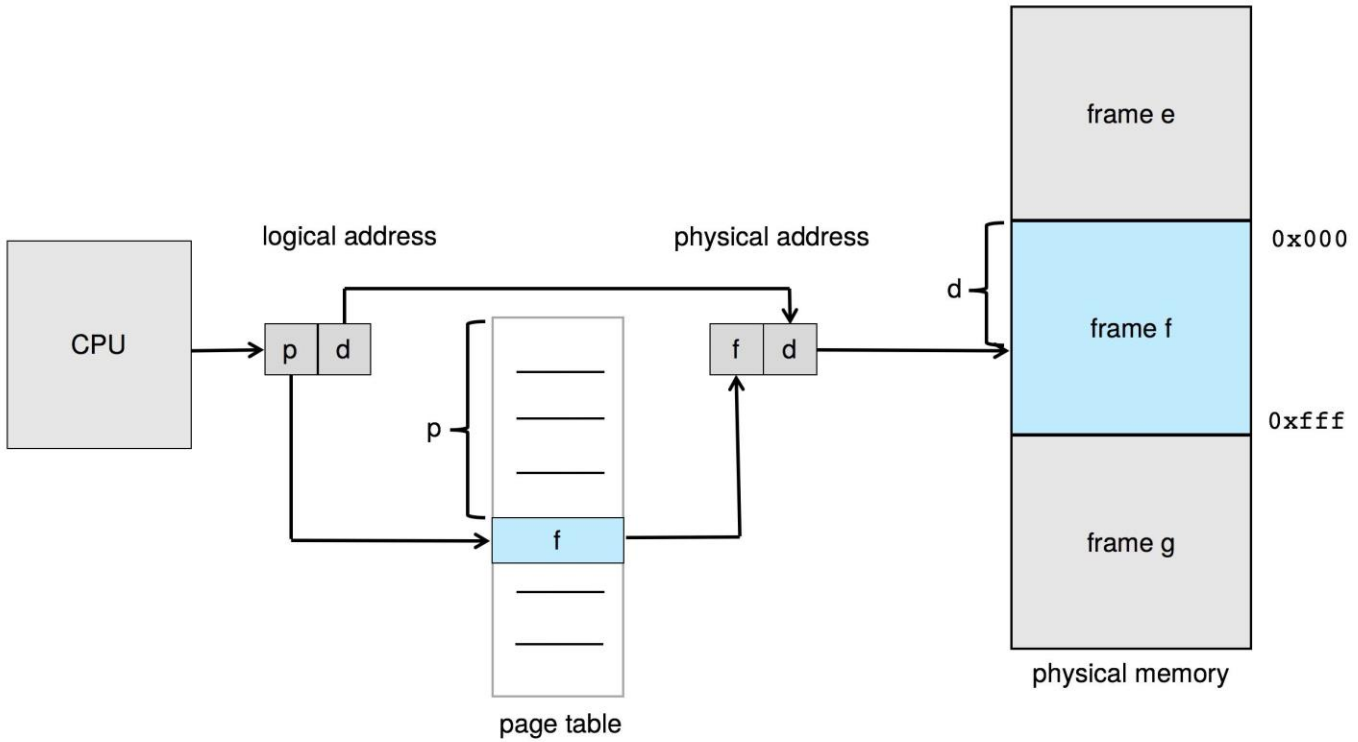
- Address generated by CPU is divided into:
  - **Page number** (***p***) – used as an index into a **page table** which contains base address of each page in physical memory.
  - **Page offset** (***d***) – combined with base address to define the physical memory address that is sent to the memory unit.



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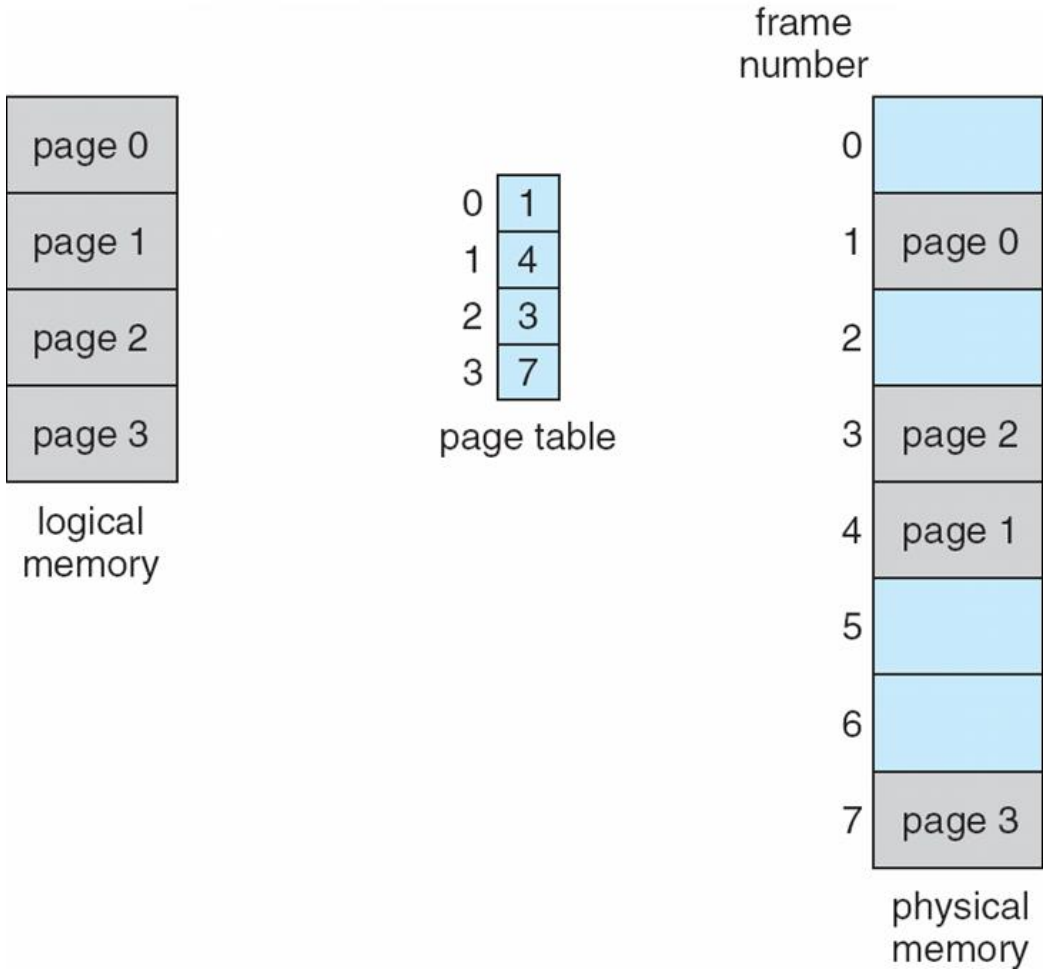
Paging Hardware





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Paging Model of Logical and Physical Memory



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